

to then allow the programmers of content to script smells into the mix as well. In demos they've shown people some roses, which you can pick up and smell, and reports are that the smell is very natural and real feeling. While it's mostly earthy and natural scents that are commonly experienced such as flowers, citrus, pine forests and the like, it's a start. And this is a product, there's plenty more being worked on in laboratories.

Another product already on sale comes from Feel Real, which is a VR headset that not only gives you smells, but also blows air at you if it senses fast movement or explosions, and the like. It uses a combination of nine individual aroma capsules to mix and create the smell of your VR world. They currently have 255 scents available.

Touch

Meta (Facebook) itself is working on, or rather, providing in house funding to researchers that are working on haptic gloves, which we have shown in the picture below. Reality Labs is like an incubator for research in the field of better simulating VR and AR, and their haptic glove is an attempt to solve touch inside a VR world. The goal here is to provide feedback to fingertips so that, let's say, a sculptor could work in the metaverse in a clay simulator, and feel his creation. It would also allow you to type on a virtual keyboard, but give your fingertips the sensation of hitting keys, instead of just flapping them about in the air.

The technology works by using many little microbubbles to inflate and deflate and provide the sensation of feeling to your fingertips. And they've developed a chip as well which they call a high-speed microfluidic processor to control and quickly

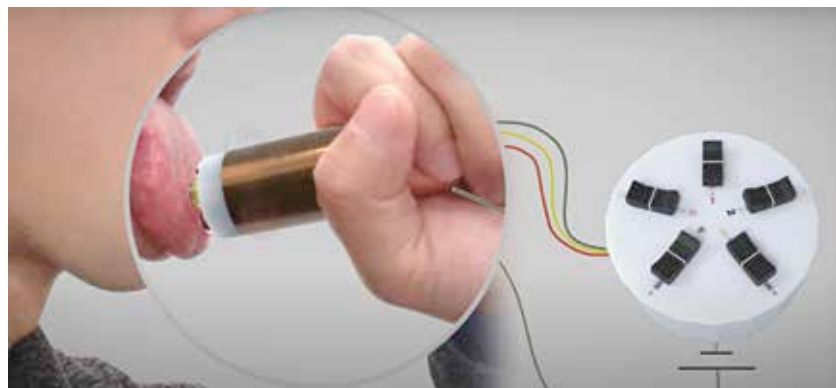
inflate and deflate the tiny microbubbles that give your fingertips the sensation. It's a super complex job tying the haptics into the visuals, so there is tonnes of physics computations happening. For example, imagine a leaking faucet, which you see in VR. You take your finger and place it under the faucet and the first drop misses, you adjust your finger and the second drop hits the right side of your finger, you adjust slightly again, and the third drop hits the middle of the finger. The first one is simple, do nothing. However, the second and third drop will need a very different computation to transfer the right sensation to your finger.

Taste

There's a concept device called Taste The TV (TTTV) that's developed by Meiji University professor Homei

often forget about this and think of us humans as having only five senses, when in fact we have six. Proprioception (or kinaesthesia) is our ability to sense the body's position and movement without looking. It's why you can close your eyes and touch your nose. Alcohol (and drugs) messes with this ability of the body, and this is why traffic police will ask you to touch your nose or walk in a line while looking up or walk a few steps backwards to check if you are impaired.

This also comes into play when we are inside a metaverse, because we need to be able to sense movement. In fact, it's when senses disagree about what we're experiencing that feelings of nausea arise. Most commonly, a visual and auditory sensation on movement while the body is sitting still causes us to feel sick. This is why



Does it taste the same?

Source: Miyashita Lab via YouTube

Miyashita. What this device does is spray a set of complex flavouring agents onto a screen that can be licked off to mimic the flavour of something displayed on it. The idea is essentially the same as what people are working on for smell, which is to narrow down smell and taste to a fixed set of fundamental ingredients, and then mix and match to change tastes or smells. There will be increased spending for more research and development now thanks to Meta pushing the metaverse concept to everyone, and we're bound to hear of new breakthroughs in these fields.

Movement

One of the hardest things to mimic is the sensation of movement. We

so many suffer from car sickness on long journeys. It's because there is not too much acceleration to remind us we're moving, but the sound and sight of us moving at high speed causes mental dissonance, and makes us dizzy. This is a problem we have solved with complex gyroscopes, usually used in military or astronaut training, but it's not something that's ready for your living room or gaming room just yet.

Will any of these technologies develop fast enough to make the metaverse more real, or will the metaverse remain just a VR gimmick that a lot of us use from time to time, but don't really "live" inside of? That is the ten trillion dollar question, isn't it? **d**



Source: Meta